

Inverse Reinforcement Learning Approach for Elicitation of Preferences in Multi-objective Sequential Optimization

1st Akiko Ikenaga

Dept. of Urban Environment Systems
Chiba University
Chiba, Japan
a.ikenaga@chiba-u.jp

2nd Sachiyo Arai

Dept. of Urban Environment Systems
Chiba University
Chiba, Japan
arai@tu.chiba-u.ac.jp

Abstract—It is crucial to know which criterion should be focused on, in a multi-objective decision making context, to select the best alternative from the multiple Pareto optimal solutions. However, in general, it is hard for the decision maker to express his/her own preference order for each criterion. In this study, we propose a preference elicitation method to estimate relative importance in terms of weights for each criterion by observing his/her processes of decision making. This method would make expert's preference elicited, and contribute at an important decision making point, such as urban planning.

Index Terms—Preference elicitation, Multi-objective sequential decision making, Inverse reinforcement learning.

I. INTRODUCTION

Preference elicitation has been a one of important issues in Artificial Intelligence studies which try to implement human expert's priority of criteria by means of the numerical weights. However, generally, experts cannot rank their preferences numerically. Though there exist many researches concerning preferences elicitation, they have focused on not sequential but one-shot decision making. While, our proposed method focuses on the preference elicitation in the context of multi-objective sequential decision making. In the multi-objective sequential decision making, the utility of each decision is clarified only after several sequence of decision process. The existing approaches to solving multi-objective decision making and multi-objective sequential decision making can be classified into two classes depending on whether *weights for each criterion* w are given or not (Table I). When w are not given, there is a phase in which the decision makers have to select the solution suitable for their preference. However, even in the methods of giving weights, little attention has been given to the means of assigning the weights for each criterion.

The proposed method quantifies preferences by putting the numerical weights for each criterion, for multi-objective sequential decision making. The contribution of the proposed method has two points of usefulness and comparability. First, this eliciting contributes to the construction of expert systems in multi-objective sequential decision making by quantifying the preferences of the experts in target problem. Second, the proposed method contributes to planning strategies by

analyzing individual's preferences. We applied the proposed method to the Deep Sea Treasure, which is a typical benchmark environment to evaluate the effectiveness of it. The proposed method could be applicable to practical problems, such as water supply problem [5] and also to a kind of logistic planning problems in which required sequential decisions.

Table I
TAXONOMY OF APPROACHES TO MULTI-OBJECTIVE DECISION MAKING

	Given weights	Not given weights
Multi-objective decision making	Multi-attribute utility function method [1]	AHP [2]
Multi-objective sequential decision making	Chebyshev scalarization [3]	Convex hull value iteration [4]

Table II
DEFINITION

$\mathcal{S}$	Set of states	$\mathcal{P}$	Set of state transition probabilities	π	Policy
$\mathcal{A}$	Set of actions	γ	Discount factor	μ	Feature expectations
$\mathcal{R}$	Reward function	m	Number of objectives	t	Finish determination value
$\mathcal{R} = (R_1(s, a), \dots, R_n(s, a))$			Reward vector	i	Iteration number

II. PROBLEM DOMAIN

Sequential decision making is modeled in Markov decision process (MDP) $\langle \mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{P}, \gamma \rangle$ (Table II). MDP is a mathematical model for dynamic optimization of stochastic systems with Markov property in state transition. Markov property refers to the property in which the next state s_{t+1} is determined only by the current observation state s_t and action a_t . The MDP is generally assumed in reinforcement learning and guarantees that reinforcement learning obtains optimal policies. The multi-objective Markov decision process (MOMDP) $\langle \mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{P}, \gamma \rangle$ is an environment that extends the MDP to multi-objective problems [6]. Multi-objective sequential decision making is modeled in MOMDP.

We show two scenarios to select single solution for multi-objective sequential decision making (Figure 1) [7]. In the first scenario, w is given and thus scalarizing multi-objective to single objective is possible at the time of planning or learning. In the second scenario, w is not given and thus a set of pareto optimal solutions are found by planning or learning. Then the decision makers select single solution based on their

preference. The pareto optimal is a state that is forced to degrade the value of at least another one objective function to improve the value of an objective function. Multi-objective reinforcement learning generally assumes the MOMDP and aims to obtain a set of pareto optimal policies.

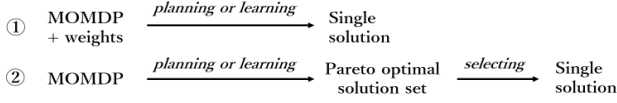


Fig. 1. Two scenarios

III. PREFERENCE ELICITATION METHOD

Our preference elicitation method is based on *Apprenticeship Learning* [8] which is one of inverse reinforcement learning (Table III). In the proposed method, an expert is defined as “the decision maker for the given action trajectory.” Action trajectory refers to the sequence of decision making. We propose a method to estimate the weights for each criterion of the expert w_E given the action trajectory of the expert $\tau_E \sim \pi_E^*$. It’s notable that we change the state representation which is a component of the reward function R from feature vector ϕ to the reward vector $\mathbf{R}$. Inner product for determining the reward function $R = \mathbf{w} \cdot \mathbf{R}$ matches the scalarization of $\mathbf{R}$ by $\mathbf{w}$. Therefore, in the proposed method $\mathbf{w}$ is outputted as the weights for each criterion.

Algorithm 1 Preference Elicitation

- 1: **Given** : MOMDP $\langle S, A, T, \gamma, \mathbf{R} \rangle$, $\tau_E \sim \pi_E^*$
- 2: Compute the expert’s feature expectations μ_E
- 3: Set randomly $\mathbf{w}^{(0)}$ in $[-1, 1]^k$ and $\|\mathbf{w}^{(0)}\| = 1$
Using the RL algorithm, compute the optimal policy $\pi^{(0)}$ for the MDP using reward $R = (\mathbf{w}^{(0)})^T \mathbf{R}$
Compute $\mu^{(0)} = \mu(\pi^{(0)})$, and set $i = 1$
- 4: Projection method
- 5: If $t^{(i)} \leq \epsilon$, then terminate
- 6: If not $t^{(i)} \leq \epsilon$ consecutively, then return to step 3
- 7: Using the RL algorithm, compute the optimal policy $\pi^{(i)}$ for the MDP using reward $R = (\mathbf{w}^{(i)})^T \mathbf{R}$
- 8: Compute $\mu^{(i)} = \mu(\pi^{(i)})$
- 9: Set $i = i + 1$, and return to step 4
- 10: Compute $w_i = w_i / \sum_{i=1}^m |w_i|$
- 11: **Return** : $\mathbf{w}$

Table III
APPRENTICESHIP LEARNING VS. PROPOSED METHOD

	Apprenticeship Learning	Proposed Method
Given	$\langle S, \mathcal{A}, \mathcal{P}, \gamma \rangle$	$\langle S, \mathcal{A}, \mathbf{R}, \mathcal{P}, \gamma \rangle$
Reward function	$R = \mathbf{w} \cdot \phi$	$R = \mathbf{w} \cdot \mathbf{R}$
Output	R	$\mathbf{w}$

IV. EXPERIMENT

The proposed method is evaluated by applying it to the Deep Sea Treasure (DST) environment (Figure 2), which is a well-known multi-objective reinforcement learning benchmark problem [9]. The action of a submarine is to move one cell up and down, left and right. A submarine begins from the cell on

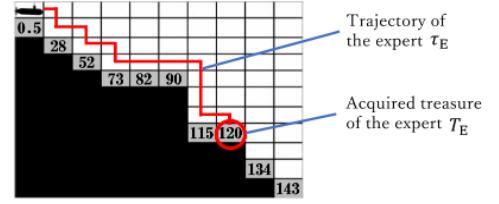


Fig. 2. Deep Sea Treasure (10 × 11 grid in which, black cells represent the seabed and gray cells represent treasure)

the upper left and when it reaches the cell with the treasure, the episode concludes. The agent manipulates the submarine and learns the policy for the two objectives of minimizing the number of steps to the treasure and maximizing the value of acquired treasure. The first element of the reward vector is a time penalty that gives -1 for every step. The second is a reward by the treasure value, and when the agent reaches a cell with treasure, the value of the treasure is rewarded and otherwise 0 is given. Also, $\gamma = 0.95$. In a DST environment, each Pareto optimal policy π^* is a policy for acquiring one of 10 kinds of treasures by the shortest route, and there are 10 Pareto optimal policies. At this time, the agent’s preference can be divided into 10 types. In this study, when the action trajectory of the expert τ_E based on the Pareto optimal policy of the expert π_E^* is given, we estimate w_E corresponding to the π_E^* . Acquisition of treasure T is based on π^* corresponding to $\mathbf{w}$. In other words, π^* maps the weights $\mathbf{w}$ to the acquired treasure T . Based on this observation, we define that $\mathbf{w}$ was correctly estimated as w_E when “the treasure T_E that the expert obtained under w_E matches the treasure T that is obtained under $\mathbf{w}$ estimated in the experiment.”

As a treasure acquired by expert T_E , all 10 kinds of treasures were deemed conceivable, and experiments were conducted based on the action trajectory used to acquire the treasure. The experimental results demonstrated that the weights of the expert w_E can be estimated correctly, even when the action trajectory of acquiring arbitrary treasure is considered as an expert’s action trajectory.

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